

INTERESTS I am a first-year Ph.D. student in Computer Science at Purdue University, advised by Prof. Jason Wu. Combining my background in design with computational methods, I design AI interactions for designers, focusing on *agentic systems*, *collaborative sensemaking*, and *generative interfaces* for creative work. Recently, I have been bringing machine learning into the interaction loop so that AI can adapt to designers’ workflows.

EDUCATION	Purdue University, West Lafayette, IN	Sep 2026 - Present
	· Ph.D. in Computer Science · Advisor: Prof. Jason Wu	
	Korea Advanced Institute of Science and Technology, Daejeon, Korea	2020 - Feb 2026
	· B.S. in Computer Science and Industrial Design · Summa Cum Laude (Ranked 3rd, GPA: 4.08/4.3)	
	New York University, New York, NY	Spring 2024
	· KAIST–NYU Minor in Integrated Design & Media (GPA: 4.0/4.0)	

PUBLICATIONS

CONFERENCE PAPERS

[C1] **VIVID: Human-AI Collaborative Authoring of Vicarious Dialogues from Lecture Videos**
Sulgi Choi, **Hyewon Lee**, Yoonjoo Lee, Juho Kim
ACM CHI 2024

POSTERS AND DEMOS

[P4] **Hangulo: Demonstrating Workflow-Embedded AI Support for Korean Lettering Implementation**
Hyewon Lee, Tak Yeon Lee
ACM UIST Adjunct 2026

[P3] **CORAL: Continually Learning Generative Interfaces from Collective User Interactions**
Daniel Lee, DaEun Choi, **Hyewon Lee**, Haijun Xia
ACM UIST Adjunct 2026

[P2] **RADI: A Design Framework for Relational and Adaptive Disclosure Interfaces**
Hyewon Lee*, Ihchae Ryu*, Yumin Cho*, Hyunseung Lim, Hwajung Hong
ACM UIST Adjunct 2025

[P1] **CamARa: Exploring and Creating Camera Movements with Spatial Reference in Augmented Reality**
Hyewon Lee, Christopher Bannon, Andrea Bianchi
ACM CHI Extended Abstracts 2025

PREPRINTS AND ARCHIVES

[A3] **GUIDE: Designer-in-the-loop Authoring of Conformant Generative User Interfaces**
Hyewon Lee, Ziyang Wang, Aiden Moy, Saran Nagubandi, Jason Wu
arXiv 2026

[A2] **“When to Hand Off, When to Work Together”: Understanding Concurrent Human-Agent Interaction in Shared Co-Creative Workspaces**

Kihoon Son, **Hyewon Lee**, DaEun Choi, Yoonsu Kim, Tae Soo Kim, Yoonjoo Lee, John Joon Young Chung, HyunJoon Jung, Juho Kim
arXiv 2026

[A1] **An Augmented Reality Camera Layout System for Intuitive, Spatially Aware Shot Design**

Hyewon Lee, Jee Youn Kim, Christopher Bannon, Andrea Bianchi
Preprint 2025

**RESEARCH
EXPERIENCE**

KIXLAB, KAIST Daejeon, Korea
Research intern

Fall 2023; Jul 2025 – Aug 2026

- Advisor: **Prof. Juho Kim**
- Researched human–AI co-authoring of pedagogical dialogues for instructor support [C1].
- Exploring synchronous co-design interactions with in-tool generative agents [A2].

Graduation Research, KAIST Daejeon, Korea
Undergraduate researcher

Mar 2025 – Dec 2025

- Advisor: **Prof. Tak Yeon Lee**
- Investigating human–AI collaborative vector editing workflows for Korean lettering through designer-centered sensemaking artifacts [P4].

MakeLab, KAIST Daejeon, Korea
Research intern

Aug 2024 – Jun 2025

- Advisor: **Prof. Andrea Bianchi**
- Developed an AR-based camera tool bridging 3D scenes and real environments to support expressive cinematography [P1] [A1].

**WORK
EXPERIENCE**

SketchSoft Seoul, Korea
Software engineer intern

Jul 2022 – Dec 2022

- Developed a 3D sketching tool Feather (Three.js) and implemented GLTF curve transformation reducing file size by 87%.
- Proposed and designed a movie export interaction allowing users to record and share 3D sketches, extending the tool’s expressive capabilities.

SPARCS, KAIST Daejeon, Korea
Software engineer & UX designer

Mar 2021 – Dec 2024

- Designed and implemented *Biseo*, an internal web service supporting voting and decisions for KAIST Programmers Association, later adopted by two additional campus groups.
- Improved UX as a design lead by resolving key usability issues and introducing an atomic design system for scalable front-end development.

**AWARDS &
HONORS**

Korean Government Scholarship Program for Study Overseas (KGSPSO),
Government of the Republic of Korea

2026

- Awarded to support doctoral study abroad; \$40,000 scholarship.

Ross Fellowship with Herbold Scholarship, Purdue University

2026

- Awarded to incoming Ph.D. students; \$10,000 scholarship.

Next-Generation Engineer Award,

Institute for Promotion of Engineering and Science of Korea, IPESK

2025

	· Recognized as an outstanding student for excellence in research.	
	Undergraduate Research Program, KAIST	2024
	· Selected for Undergraduate Research Program (URP); \$5,000 research grant.	
	College of Engineering Dean's List, KAIST	2022
	· Achieved a perfect GPA of 4.3/4.3 (2nd in department); \$800 scholarship.	
	Lim Misuk Scholarship, KAIST	2022
	· Recognized for leadership and societal contribution; \$1,000 scholarship.	
	Presidential Science Scholarship, Ministry of Science and ICT, Republic of Korea	2020
	· Awarded by the President of Korea to outstanding science and engineering students. · Covered full tuition (4 years) and \$20,000 scholarship.	
TEACHING & MENTORING	Research Mentor, AFE	2025 - 2026
	· Mentored high school students on research projects in computer science.	
	Undergraduate Tutor, KAIST	2022 - 2023
	· Taught data structures and algorithms; nominated as Best Tutor (2023).	
	Teaching Assistant, KAIST	2021
	· Taught CS101: Introduction to Programming (Python) lab sessions.	
SERVICES & LEADERSHIP	Student Volunteer, ACM SIGCHI Korea Local Chapter	2025
	Taekwondo Club Organizer, KAIST	2024 - 2025
	International Student Buddy, KAIST	2024
	Department President, School of Computing, KAIST	2022
	Cohort Representative, School of Computing, KAIST	2021